

Salyut

Chelomei and his staff developed a military station for two to three cosmonauts, with a design life of 1 to 2 years. They designed an integrated system: a single-launch space station dubbed Almaz (“diamond”) and a Transport Logistics Spacecraft for reaching it. Chelomei’s three-stage Proton booster would launch them both.

Almaz was to be equipped with a crew capsule, radar remote-sensing apparatus for imaging the Earth’s surface, cameras, two re-entry capsules for returning data to Earth, and an anti-aircraft cannon to defend against American attack. It would be the world’s first military space station.

However, when NASA announced it was also working on a space station – Skylab – the Soviets sought to speed up the launch by taking Chelomei’s research and handing it to the group formerly headed by Korolev, given their success with the Soyuz spacecraft. The space station was renamed Salyut (“salute”) and Salyut 1 was successfully launched on April 19, 1971,



A model of the Salyut space station

becoming the first space station to orbit Earth. It used a “monolithic” design, which meant that it was completely constructed and assembled on Earth and then launched into space.

The purpose of Salyut was to test the elements of the systems of a space station and to conduct scientific research and experiments. The craft was being 20 metres in length, 4 metres in maximum diameter, and 99 cubic cm in interior space. Of its several compartments, three were pressurized (100 cubic m total), and two could be entered by the crew.